

Abstract

This thesis presents the development and characterization of the experimental subsystems required for the preparation, optical trapping, and controlled transport of cold ^{87}Rb atoms toward a Hollow-Core Photonic Crystal Fiber (HCPCF).

The first part of the work concerns the preparation of the atomic sample. The optical molasses was optimized by adjusting the cooling-laser detuning, the cooling-beam amplitude, the ramp duration, and the compensation magnetic fields. Under the optimized conditions, a minimum temperature of approximately $T = (8.4 \pm 0.2) \mu\text{K}$ was obtained from ballistic-expansion measurements. The loading of the Optical Dipole Trap (ODT) was then investigated by translating the MOT position to improve its spatial overlap with the dipole-trap beam. A loading efficiency of $(0.161 \pm 0.005) \%$ was measured, corresponding to a calibrated trapped population of approximately $(1.0 \pm 0.1) \times 10^5$ atoms.

The optical conveyor belt requires two counter-propagating 1064 nm fields. To provide the second field, an injection line for a polarization-maintaining single-mode fiber was designed, characterized, and aligned. The transmitted field was spatially overlapped with the counter-propagating HCPCF branch on the upper breadboard. The stability of the black-fiber output was characterized in both the time and frequency domains. The fractional power Allan deviation reached a minimum of approximately 8×10^{-4} at an averaging time of 0.1 s, while the angular Allan deviation of the measured polarization reached approximately 7×10^{-5} rad at an averaging time of 3 s. The measurements also revealed the presence of low-frequency technical noise and slow environmental drifts.

Because optical transport can increase the motional energy of the atoms, Electromagnetically Induced Transparency (EIT) cooling was investigated numerically. A quantum master-equation model was implemented in QuTiP, first for an ideal three-level Λ system and subsequently for a 24-level model including the hyperfine and Zeeman structure of the ^{87}Rb D_2 transition. The multilevel simulations show that optical pumping into uncoupled ground states interrupts the cooling cycle in the absence of repumping. When an active repump field is included, the EIT cooling dynamics are sustained and the simulated atomic motion approaches the motional ground state.

The results establish the main experimental and numerical prerequisites for a future implementation of conveyor-belt transport and sub-Doppler cooling of cold atoms inside the HCPCF. The complete dynamic transport sequence and its experimental validation

remain as subsequent steps.